

```
In [6]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
sns.set_theme(style="whitegrid")
```

```
In [8]: df=pd.read_csv('Amazon Sale Report.csv', sep=';')
```

Data Characteristic

```
In [9]: df.head()
```

```
Out[9]:
```

	index,Order ID,Date,Status,Fulfilment,Sales Channel ,ship-service-level,Style,SKU,Category,Size,ASIN,Courier Status,Qty,currency,Amount,ship-city,ship-state,ship-postal-code,ship-country,promotion-ids,B2B,fulfilled-by,Unnamed: 22
0	0,405-8078784-5731545,04-30-22,Cancelled,Merch...
1	1,171-9198151-1101146,04-30-22,Shipped - Deliv...
2	2,404-0687676-7273146,04-30-22,Shipped,Amazon,...
3	3,403-9615377-8133951,04-30-22,Cancelled,Merch...
4	4,407-1069790-7240320,04-30-22,Shipped,Amazon,...

```
In [4]: df.tail()
```

Out[4]:

	index	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	...	currency	Amount	ship-
128970	128970	406- 6001380- 7673107	05- 31- 22	Shipped	Amazon	Amazon.in	Expedited	JNE3697	JNE3697- KR-XL	kurta	...	INR	517.0	HYDERA
128971	128971	402- 9551604- 7544318	05- 31- 22	Shipped	Amazon	Amazon.in	Expedited	SET401	SET401- KR-NP- M	Set	...	INR	999.0	GURUGI
128972	128972	407- 9547469- 3152358	05- 31- 22	Shipped	Amazon	Amazon.in	Expedited	J0157	J0157- DR-XXL	Western Dress	...	INR	690.0	HYDERA
128973	128973	402- 6184140- 0545956	05- 31- 22	Shipped	Amazon	Amazon.in	Expedited	J0012	J0012- SKD-XS	Set	...	INR	1199.0	F
128974	128974	408- 7436540- 8728312	05- 31- 22	Shipped	Amazon	Amazon.in	Expedited	J0003	J0003- SET-S	Set	...	INR	696.0	Ra

5 rows × 24 columns



In [5]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 128975 entries, 0 to 128974
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   index                 128975 non-null  int64
1   Order ID             128975 non-null  object
2   Date                 128975 non-null  object
3   Status               128975 non-null  object
4   Fulfilment           128975 non-null  object
5   Sales Channel        128975 non-null  object
6   ship-service-level   128975 non-null  object
7   Style                128975 non-null  object
8   SKU                  128975 non-null  object
9   Category             128975 non-null  object
10  Size                 128975 non-null  object
11  ASIN                 128975 non-null  object
12  Courier Status       122103 non-null  object
13  Qty                  128975 non-null  int64
14  currency             121180 non-null  object
15  Amount              121180 non-null  float64
16  ship-city            128942 non-null  object
17  ship-state           128942 non-null  object
18  ship-postal-code     128942 non-null  float64
19  ship-country         128942 non-null  object
20  promotion-ids        79822 non-null  object
21  B2B                  128975 non-null  bool
22  fulfilled-by         39277 non-null  object
23  Unnamed: 22          79925 non-null  object
dtypes: bool(1), float64(2), int64(2), object(19)
memory usage: 22.8+ MB

```

```
In [6]: df.columns
```

```

Out[6]: Index(['index', 'Order ID', 'Date', 'Status', 'Fulfilment', 'Sales Channel ',
              'ship-service-level', 'Style', 'SKU', 'Category', 'Size', 'ASIN',
              'Courier Status', 'Qty', 'currency', 'Amount', 'ship-city',
              'ship-state', 'ship-postal-code', 'ship-country', 'promotion-ids',
              'B2B', 'fulfilled-by', 'Unnamed: 22'],
              dtype='object')

```

Category: Type of product. (String) Size: Size of the product.

(String) Date: Date of the sale.

(Date) Status: Status of the sale.

(String) Fulfilment: Method of fulfilment.

(String) Style: Style of the product.

(String) SKU: Stock Keeping Unit.

(String) ASIN: Amazon Standard Identification Number.

(String) Courier Status: Status of the courier.

(String) Qty: Quantity of the product.

(Integer) Amount: Amount of the sale.

(Float) B2B: Business to business sale.

(Boolean) Currency: The currency used for the sale. (String)

```
In [7]: df.describe()
```

Out[7]:

	index	Qty	Amount	ship-postal-code
count	128975.000000	128975.000000	121180.000000	128942.000000
mean	64487.000000	0.904431	648.561465	463966.236509
std	37232.019822	0.313354	281.211687	191476.764941
min	0.000000	0.000000	0.000000	110001.000000
25%	32243.500000	1.000000	449.000000	382421.000000
50%	64487.000000	1.000000	605.000000	500033.000000
75%	96730.500000	1.000000	788.000000	600024.000000
max	128974.000000	15.000000	5584.000000	989898.000000

In [8]: `df.describe(include=('O'))`

Out[8]:

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	ASIN	Courier Status	current
count	128975	128975	128975	128975	128975	128975	128975	128975	128975	128975	128975	122103	12118
unique	120378	91	13	2	2	2	1377	7195	9	11	7190	3	
top	403-4984515-8861958	05-03-22	Shipped	Amazon	Amazon.in	Expedited	JNE3797	JNE3797-KR-L	Set	M	B09SDXFFQ1	Shipped	IN
freq	12	2085	77804	89698	128851	88615	4224	773	50284	22711	773	109487	12118

In [9]: `df.shape`

```
Out[9]: (128975, 24)
```

Data Cleaning

```
In [10]: df.isnull().sum()
```

```
Out[10]: index                0
Order ID                    0
Date                        0
Status                      0
Fulfilment                  0
Sales Channel               0
ship-service-level         0
Style                       0
SKU                         0
Category                    0
Size                        0
ASIN                       0
Courier Status             6872
Qty                         0
currency                   7795
Amount                     7795
ship-city                   33
ship-state                  33
ship-postal-code            33
ship-country                33
promotion-ids              49153
B2B                         0
fulfilled-by               89698
Unnamed: 22                 49050
dtype: int64
```

```
In [11]: df.duplicated().sum()
```

```
Out[11]: np.int64(0)
```

```
In [12]: df.nunique().to_frame(name='Count of unique values')
```

Out[12]:

Count of unique values	
index	128975
Order ID	120378
Date	91
Status	13
Fulfilment	2
Sales Channel	2
ship-service-level	2
Style	1377
SKU	7195
Category	9
Size	11
ASIN	7190
Courier Status	3
Qty	10
currency	1
Amount	1410
ship-city	8955
ship-state	69
ship-postal-code	9459
ship-country	1
promotion-ids	5787
B2B	2

Count of unique values	
fulfilled-by	1
Unnamed: 22	1

```
In [13]: df.drop(columns= ['index', 'Unnamed: 22', 'fulfilled-by', 'ship-country', 'currency'], inplace = True)
```

```
In [14]: df.isnull().sum()
```

```
Out[14]: Order ID          0
Date          0
Status        0
Fulfilment    0
Sales Channel 0
ship-service-level 0
Style         0
SKU           0
Category      0
Size          0
ASIN          0
Courier Status 6872
Qty           0
Amount        7795
ship-city     33
ship-state    33
ship-postal-code 33
promotion-ids 49153
B2B           0
dtype: int64
```

```
In [15]: df[df.duplicated(['Order ID', 'ASIN'], keep=False)]
```

Out[15]:

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	ASIN	Courier Status	Qty	#
30660	406-0372545-6086735	04-12-22	Cancelled	Amazon	Amazon.in	Expedited	SET197	SET197-KR-NP-L	Set	L	B08B3YPD63	Cancelled	0	
30661	406-0372545-6086735	04-12-22	Cancelled	Amazon	Amazon.in	Expedited	SET197	SET197-KR-NP-L	Set	L	B08B3YPD63	Cancelled	0	
41291	408-0373839-4433120	04-05-22	Cancelled	Amazon	Amazon.in	Expedited	JNE3501	JNE3501-KR-M	kurta	M	B08MYVCDB4	Cancelled	0	
41292	408-0373839-4433120	04-05-22	Cancelled	Amazon	Amazon.in	Expedited	JNE3501	JNE3501-KR-M	kurta	M	B08MYVCDB4	Cancelled	0	
52330	407-8364731-6449117	05-29-22	Shipped	Amazon	Amazon.in	Expedited	JNE3769	JNE3769-KR-L	kurta	L	B09K3T675B	Cancelled	0	
52333	407-8364731-6449117	05-29-22	Shipped	Amazon	Amazon.in	Expedited	JNE3769	JNE3769-KR-L	kurta	L	B09K3T675B	Shipped	1	
79844	171-9628368-5329958	05-07-22	Cancelled	Amazon	Amazon.in	Expedited	J0329	J0329-KR-L	kurta	L	B09KXRB7JV	Cancelled	0	
79845	171-9628368-5329958	05-07-22	Cancelled	Amazon	Amazon.in	Expedited	J0329	J0329-KR-L	kurta	L	B09KXRB7JV	Cancelled	0	

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	ASIN	Courier Status	Qty	/
	85790	171-3249942-2207542	05-03-22	Shipped	Amazon	Amazon.in	Expedited	SET323	SET323-KR-NP-XL	Set	XL	B09NDKKM7P	Shipped	1
	85791	171-3249942-2207542	05-03-22	Shipped	Amazon	Amazon.in	Expedited	SET323	SET323-KR-NP-XL	Set	XL	B09NDKKM7P	Shipped	1
	86418	405-8669298-3850736	05-03-22	Shipped	Amazon	Amazon.in	Expedited	MEN5025	MEN5025-KR-XXXL	kurta	3XL	B08YYQS8FH	Shipped	1
	86419	405-8669298-3850736	05-03-22	Shipped	Amazon	Amazon.in	Expedited	MEN5025	MEN5025-KR-XXXL	kurta	3XL	B08YYQS8FH	Shipped	1
	98954	407-4853873-4978725	06-22-22	Shipped	Amazon	Amazon.in	Expedited	J0230	J0230-SKD-M	Set	M	B08XNJG8B1	Shipped	1

	Order ID	Date	Status	Fulfilment	Sales Channel	ship-service-level	Style	SKU	Category	Size	ASIN	Courier Status	Qty	
	407- 98955 4853873-4978725	06- 22- 22	Shipped	Amazon	Amazon.in	Expedited	J0230	J0230-SKD-M	Set	M	B08XNJG8B1	Shipped	1	

```
In [16]: df.drop_duplicates(['Order ID','ASIN'],inplace = True,ignore_index=True)
```

```
In [17]: df.isnull().sum()
```

```
Out[17]: Order ID          0
Date          0
Status        0
Fulfilment    0
Sales Channel  0
ship-service-level  0
Style         0
SKU          0
Category      0
Size         0
ASIN         0
Courier Status 6872
Qty          0
Amount       7792
ship-city     33
ship-state    33
ship-postal-code 33
promotion-ids 49150
B2B          0
dtype: int64
```

```
In [18]: df["Courier Status"]
```

```
Out[18]: 0      NaN
          1    Shipped
          2    Shipped
          3      NaN
          4    Shipped
          ...
        128963  Shipped
        128964  Shipped
        128965  Shipped
        128966  Shipped
        128967  Shipped
        Name: Courier Status, Length: 128968, dtype: object
```

```
In [19]: df['Courier Status'].fillna('unknown',inplace=True) # unknown is used for str and 0 is for numerical
```

```
In [20]: df["Amount"]
```

```
Out[20]: 0      647.62
          1    406.00
          2    329.00
          3    753.33
          4    574.00
          ...
        128963  517.00
        128964  999.00
        128965  690.00
        128966  1199.00
        128967  696.00
        Name: Amount, Length: 128968, dtype: float64
```

```
In [21]: df['Amount'].fillna(0,inplace=True)
```

```
In [22]: df["promotion-ids"]
```

```
Out[22]: 0      NaN
        1      Amazon PLCC Free-Financing Universal Merchant ...
        2      IN Core Free Shipping 2015/04/08 23-48-5-108
        3      NaN
        4      NaN
        ...
        128963      NaN
        128964      IN Core Free Shipping 2015/04/08 23-48-5-108
        128965      NaN
        128966      IN Core Free Shipping 2015/04/08 23-48-5-108
        128967      IN Core Free Shipping 2015/04/08 23-48-5-108
        Name: promotion-ids, Length: 128968, dtype: object
```

```
In [23]: df['promotion-ids'].fillna('no',inplace=True)
```

```
In [24]: df['ship-city'].fillna('unknown', inplace = True)
        df['ship-state'].fillna('unknown', inplace = True)
        df['ship-postal-code'].fillna('unknown', inplace = True)
```

```
In [25]: df.isnull().sum()
```

```
Out[25]: Order ID      0
         Date         0
         Status       0
         Fulfilment    0
         Sales Channel 0
         ship-service-level 0
         Style        0
         SKU          0
         Category     0
         Size         0
         ASIN         0
         Courier Status 0
         Qty          0
         Amount       0
         ship-city    0
         ship-state   0
         ship-postal-code 0
         promotion-ids 0
         B2B          0
         dtype: int64
```

Data Visualization

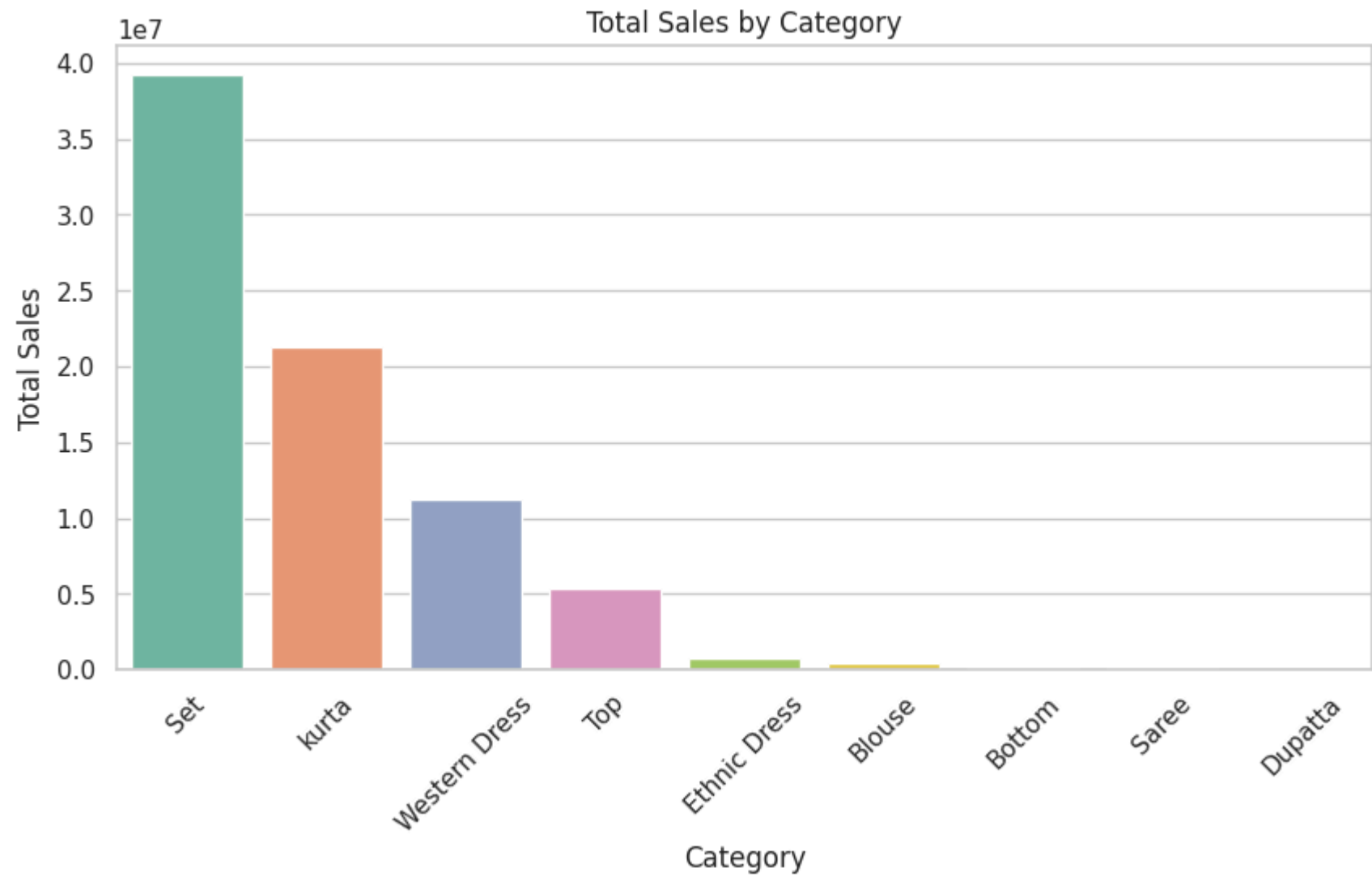
Categorical Analysis

1-Sales by Category

```
In [26]: plt.figure(figsize=(10,5))
         cat_sales = df.groupby("Category")["Amount"].sum().sort_values(ascending=False)

         sns.barplot(
             x=cat_sales.index,
             y=cat_sales.values,
             palette="Set2"
         )
```

```
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.show()
```

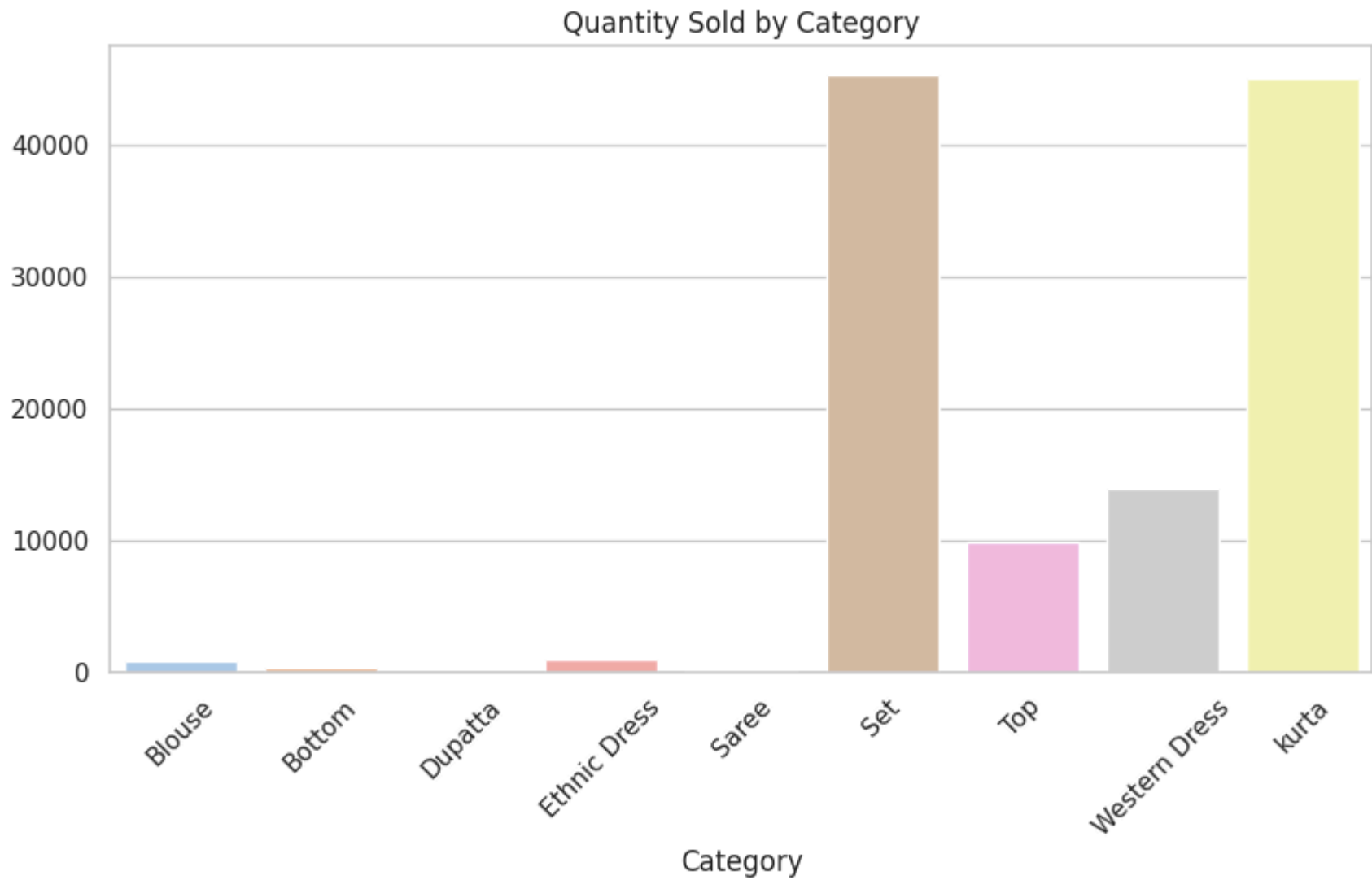


2-Quantity Sold by Category

```
In [27]: plt.figure(figsize=(10,5))
cat_qty = df.groupby("Category")["Qty"].sum()

sns.barplot(
    x=cat_qty.index,
    y=cat_qty.values,
    palette="pastel"
)

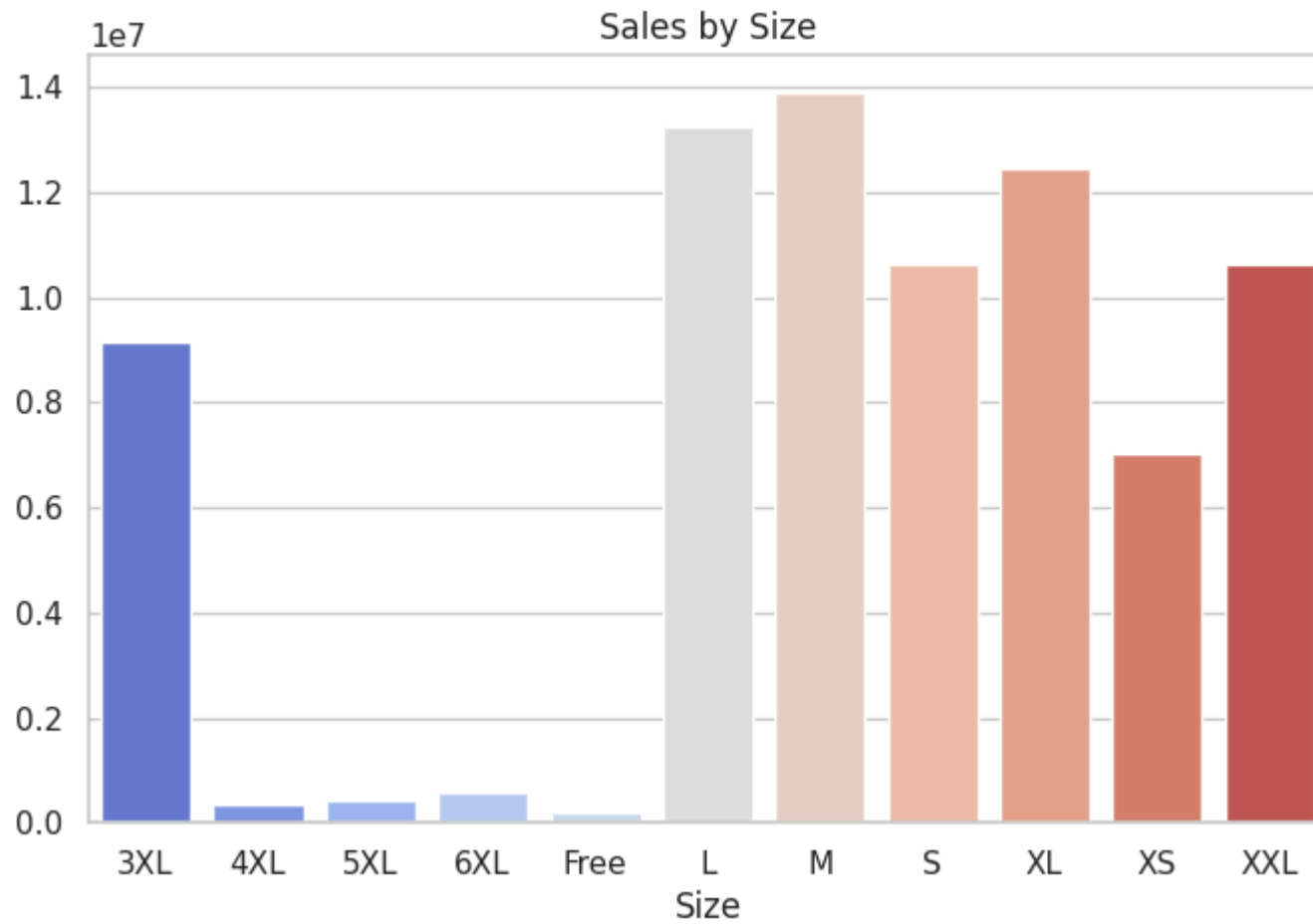
plt.title("Quantity Sold by Category")
plt.xticks(rotation=45)
plt.show()
```



Sales by Size

```
In [28]: plt.figure(figsize=(8,5))  
size_sales = df.groupby("Size")["Amount"].sum()
```

```
sns.barplot(  
    x=size_sales.index,  
    y=size_sales.values,  
    palette="coolwarm"  
)  
  
plt.title("Sales by Size")  
plt.show()
```

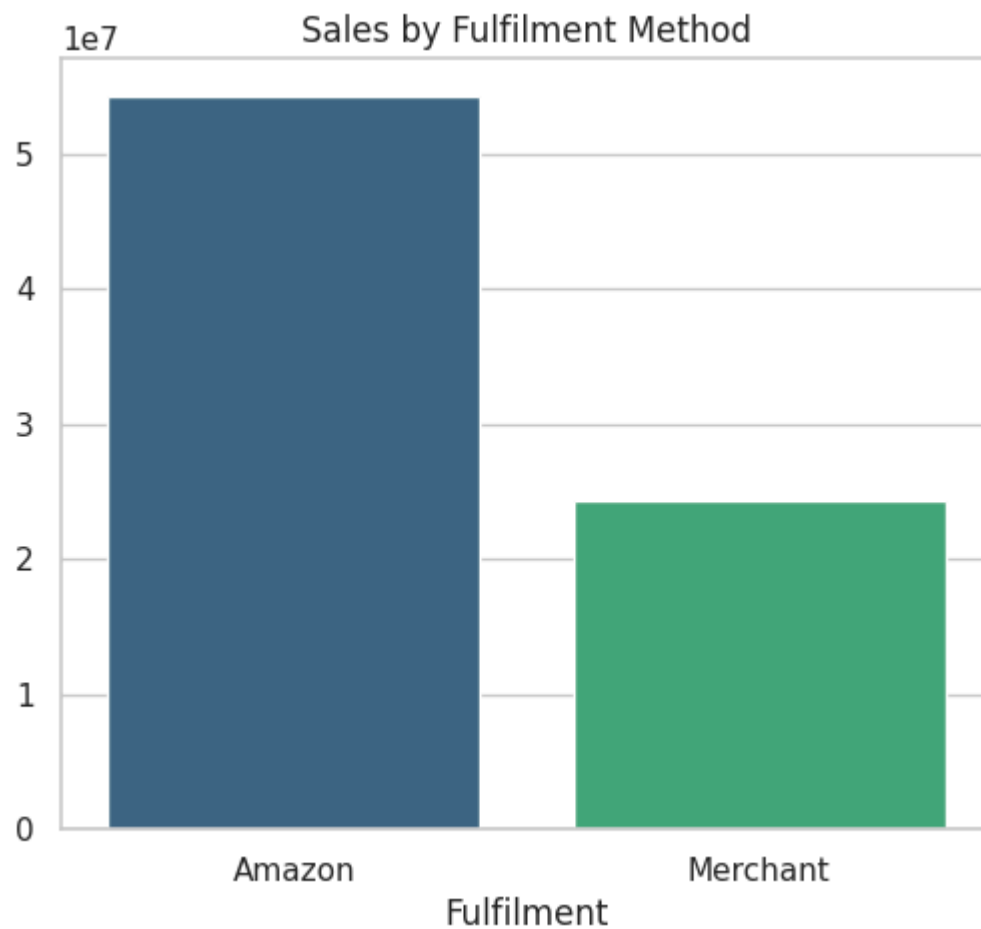


Sales by Fulfilment Method

```
In [29]: plt.figure(figsize=(6,5))
full_sales = df.groupby("Fulfilment")["Amount"].sum()

sns.barplot(
    x=full_sales.index,
    y=full_sales.values,
    palette="viridis"
)

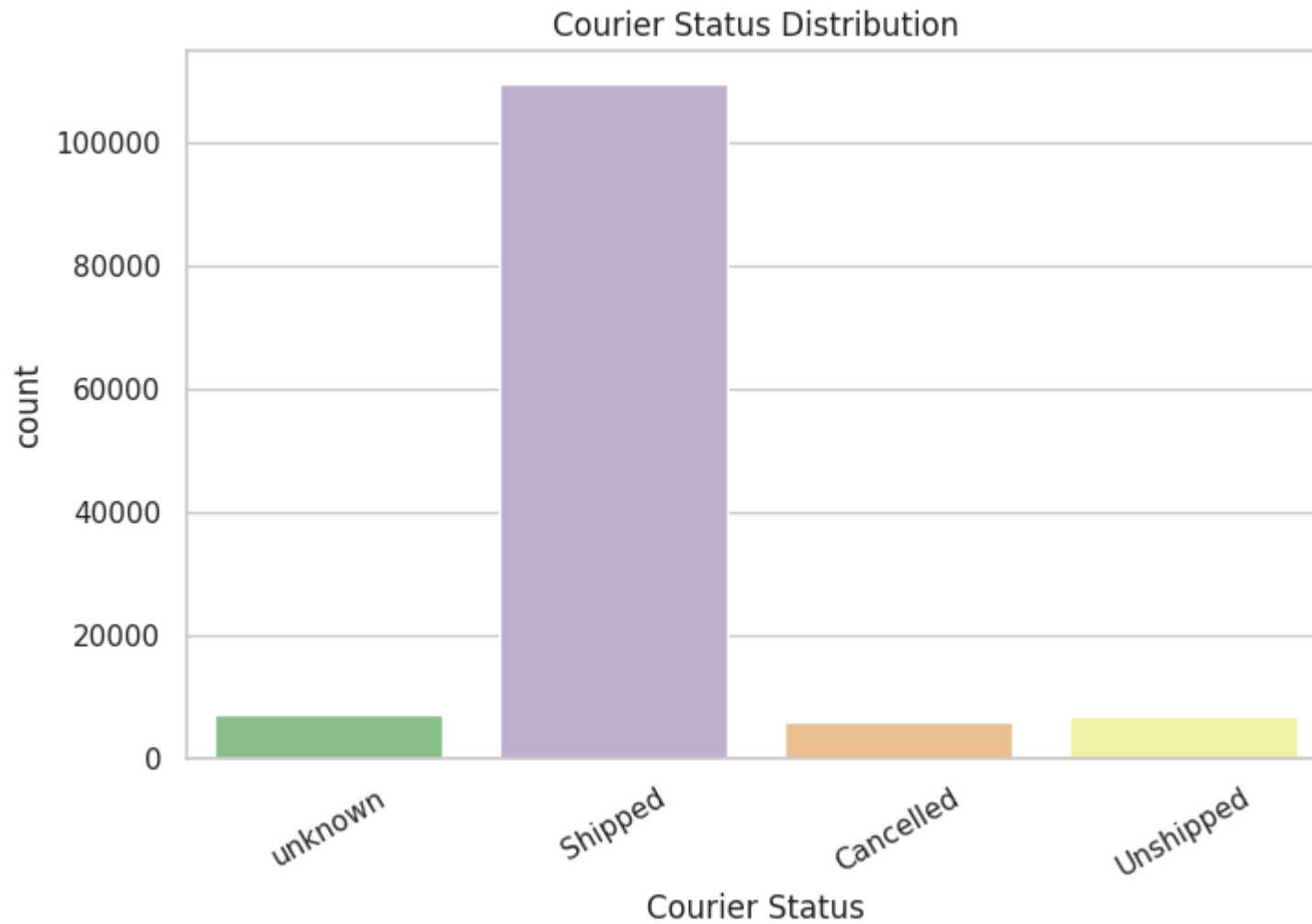
plt.title("Sales by Fulfilment Method")
plt.show()
```



Courier Status Distribution

```
In [30]: plt.figure(figsize=(8,5))
sns.countplot(
    x="Courier Status",
    data=df,
    palette="Accent"
)

plt.title("Courier Status Distribution")
plt.xticks(rotation=30)
plt.show()
```



Time Series Analysis

Sales Trend Over Time

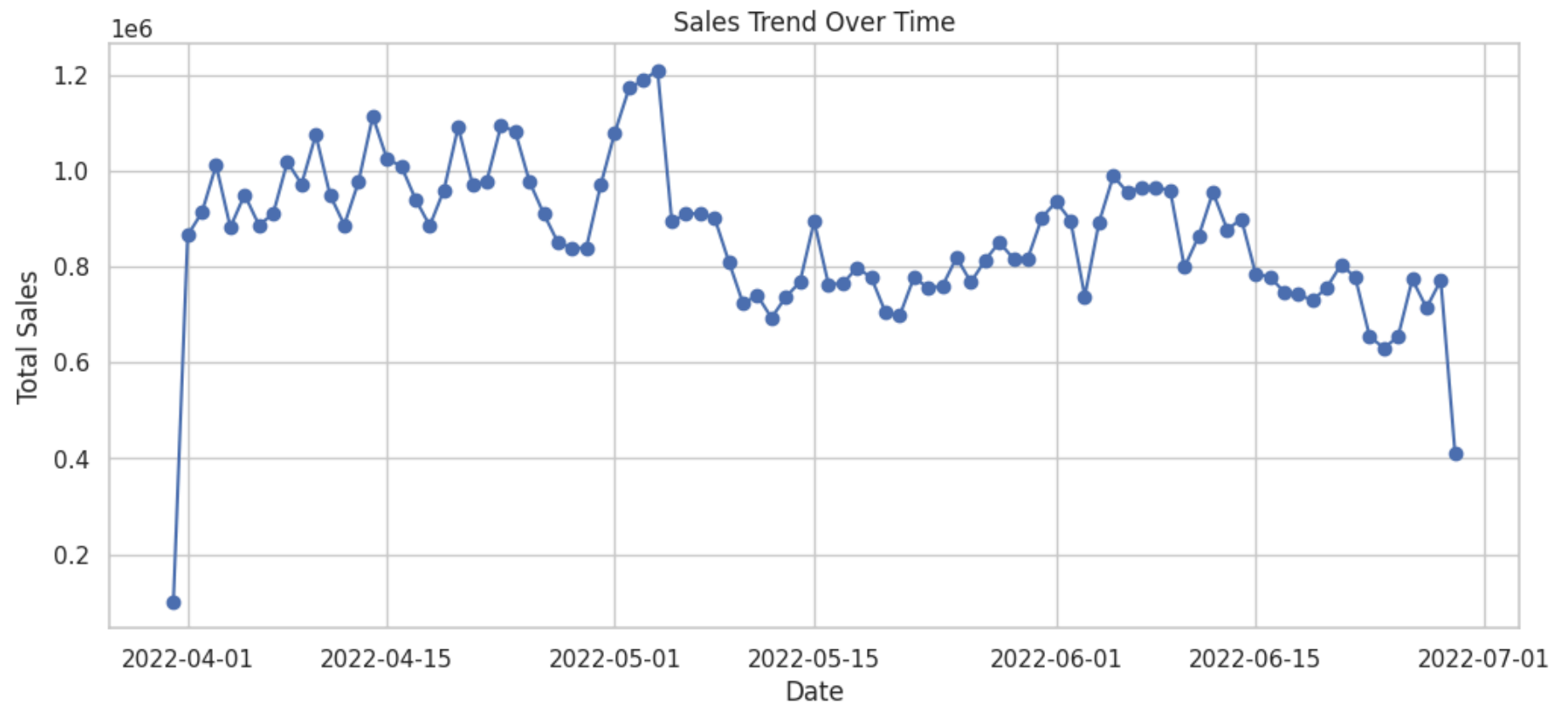
```
In [31]: df["Date"] = pd.to_datetime(df["Date"])
daily_sales = df.groupby("Date")["Amount"].sum()
```

```
plt.figure(figsize=(12,5))
plt.plot(
    daily_sales.index,
    daily_sales.values,
    marker="o"
)

plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Total Sales")
plt.show()
```

/tmp/ipykernel_17/1714131542.py:1: UserWarning: Could not infer format, so each element will be parsed individually, falling back to `dateutil`. To ensure parsing is consistent and as-expected, please specify a format.

```
df["Date"] = pd.to_datetime(df["Date"])
```

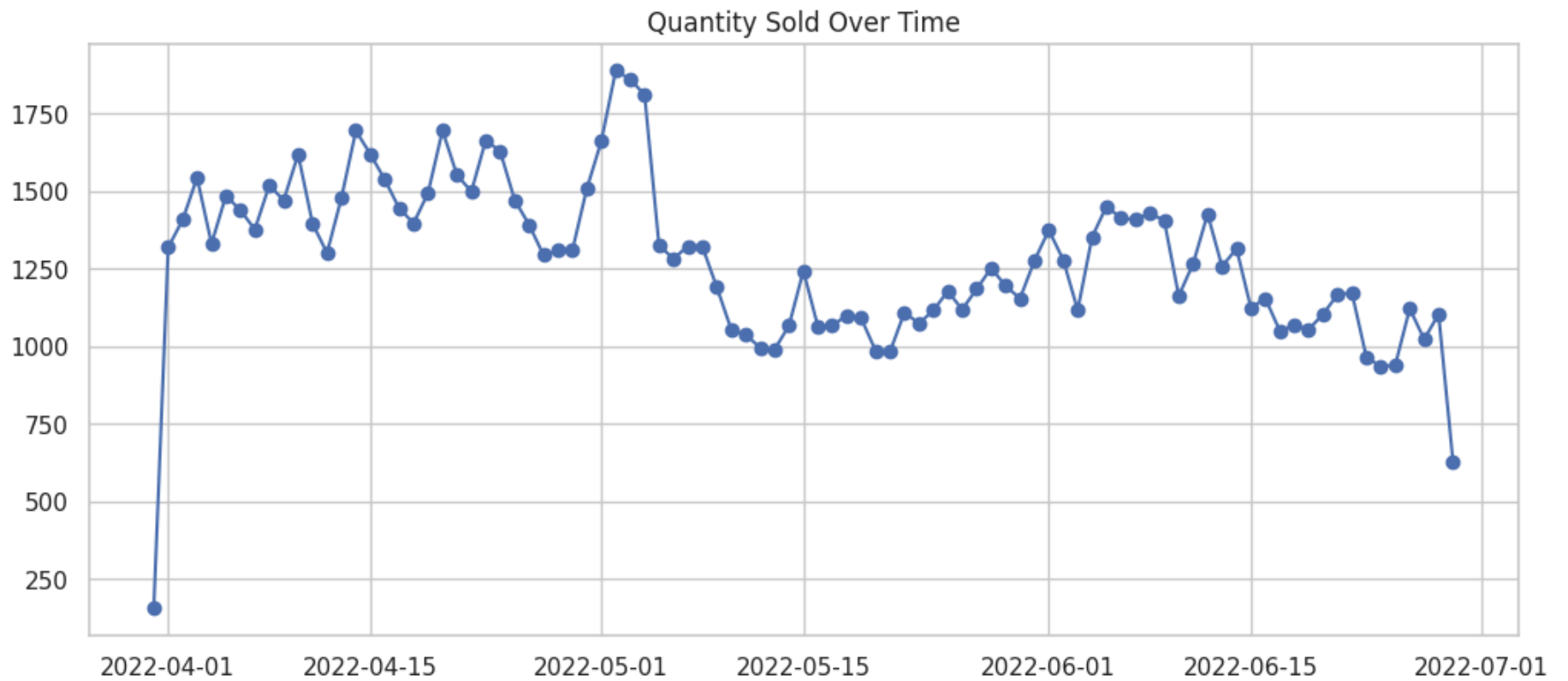


Quantity Sold Over Time

```
In [32]: daily_qty = df.groupby("Date")["Qty"].sum()

plt.figure(figsize=(12,5))
plt.plot(
    daily_qty.index,
    daily_qty.values,
    marker="o"
)

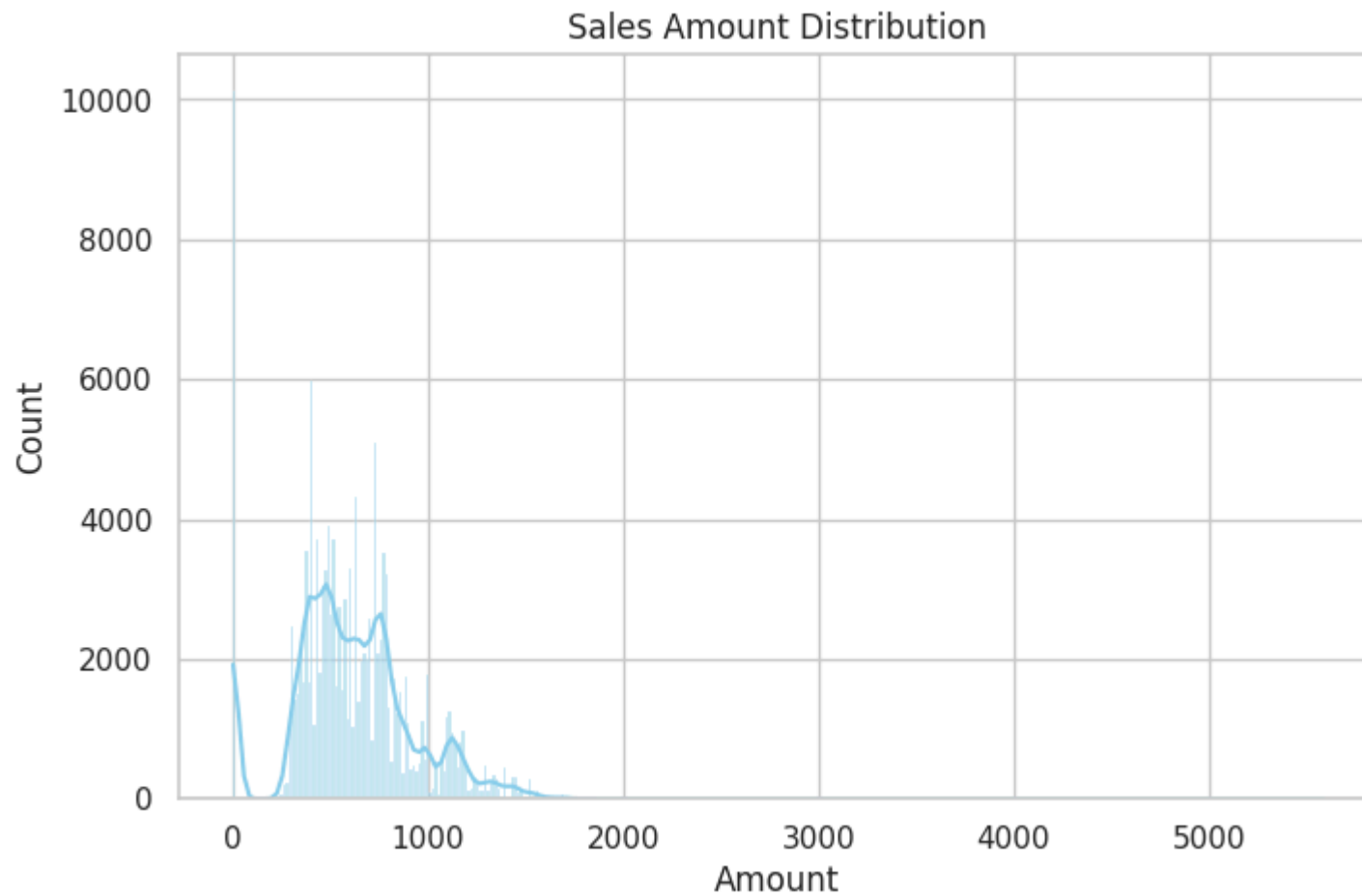
plt.title("Quantity Sold Over Time")
plt.show()
```



Distribution Analysis

```
In [33]: plt.figure(figsize=(8,5))
sns.histplot(
    df["Amount"],
    kde=True,
    color="skyblue"
)

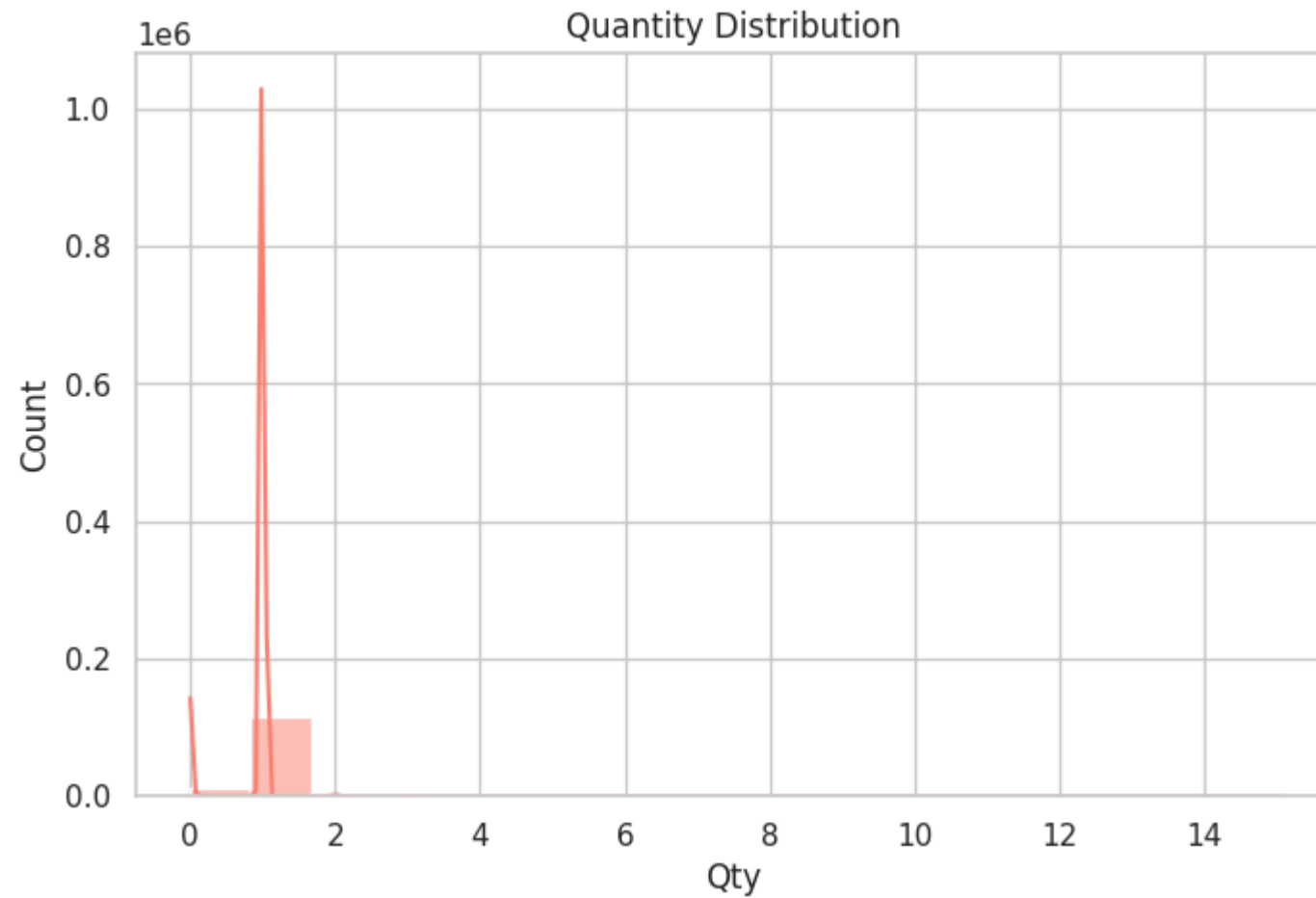
plt.title("Sales Amount Distribution")
plt.show()
```



Quantity Distribution

```
In [34]: plt.figure(figsize=(8,5))
sns.histplot(
    df["Qty"],
    kde=True,
    color="salmon"
)
```

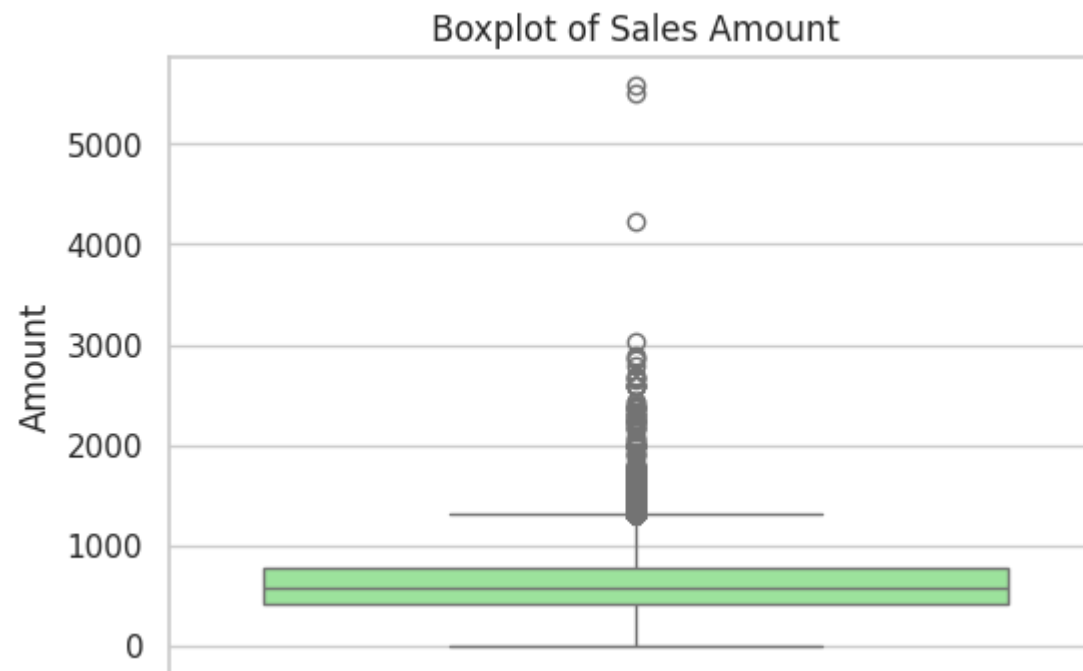
```
plt.title("Quantity Distribution")
plt.show()
```



Boxplot of Sales Amount

```
In [35]: plt.figure(figsize=(6,4))
sns.boxplot(
    y=df["Amount"],
    color="lightgreen"
)
```

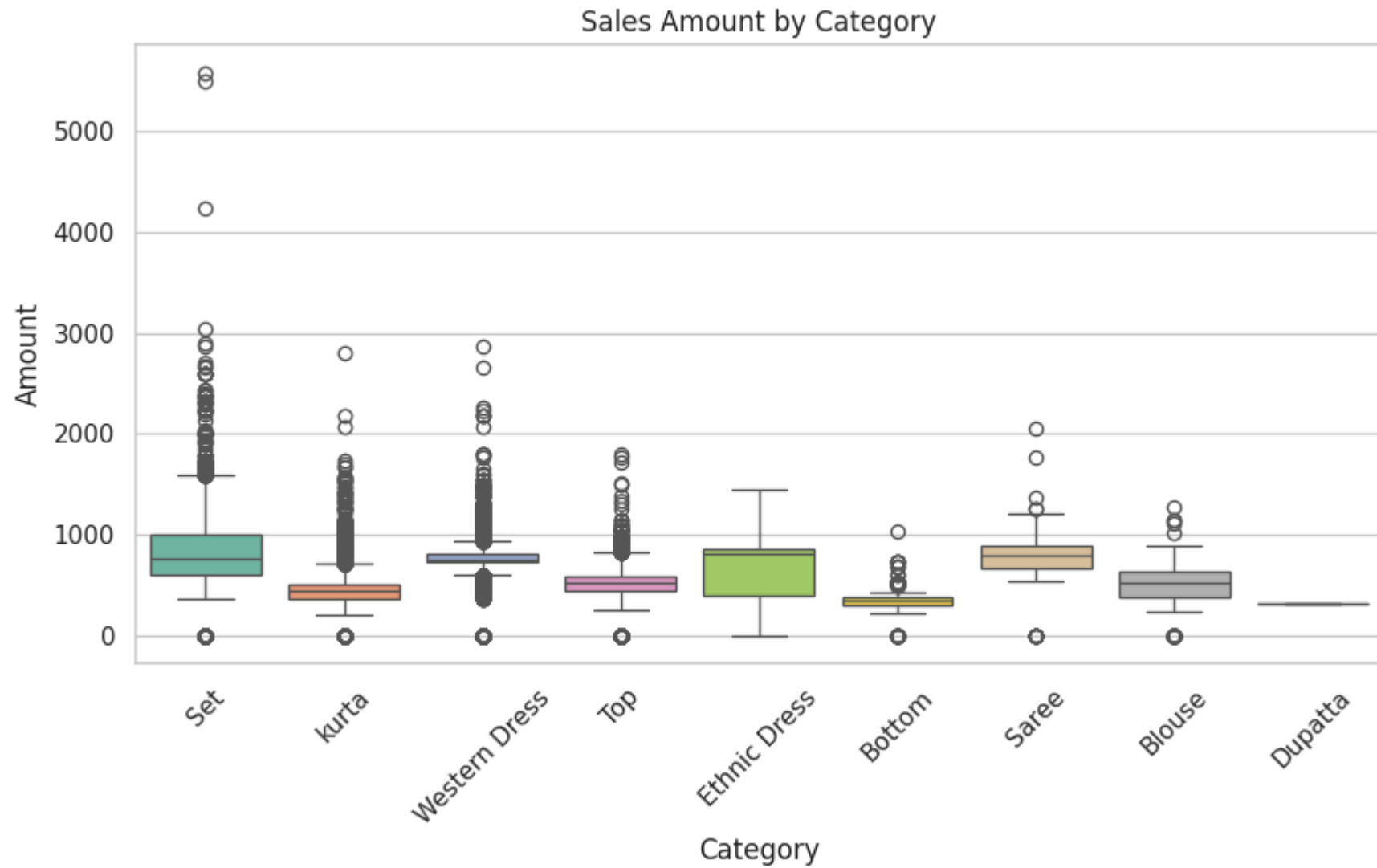
```
plt.title("Boxplot of Sales Amount")
plt.show()
```



Boxplot Amount by Category

```
In [36]: plt.figure(figsize=(10,5))
sns.boxplot(
    x="Category",
    y="Amount",
    data=df,
    palette="Set2"
)

plt.title("Sales Amount by Category")
plt.xticks(rotation=45)
plt.show()
```



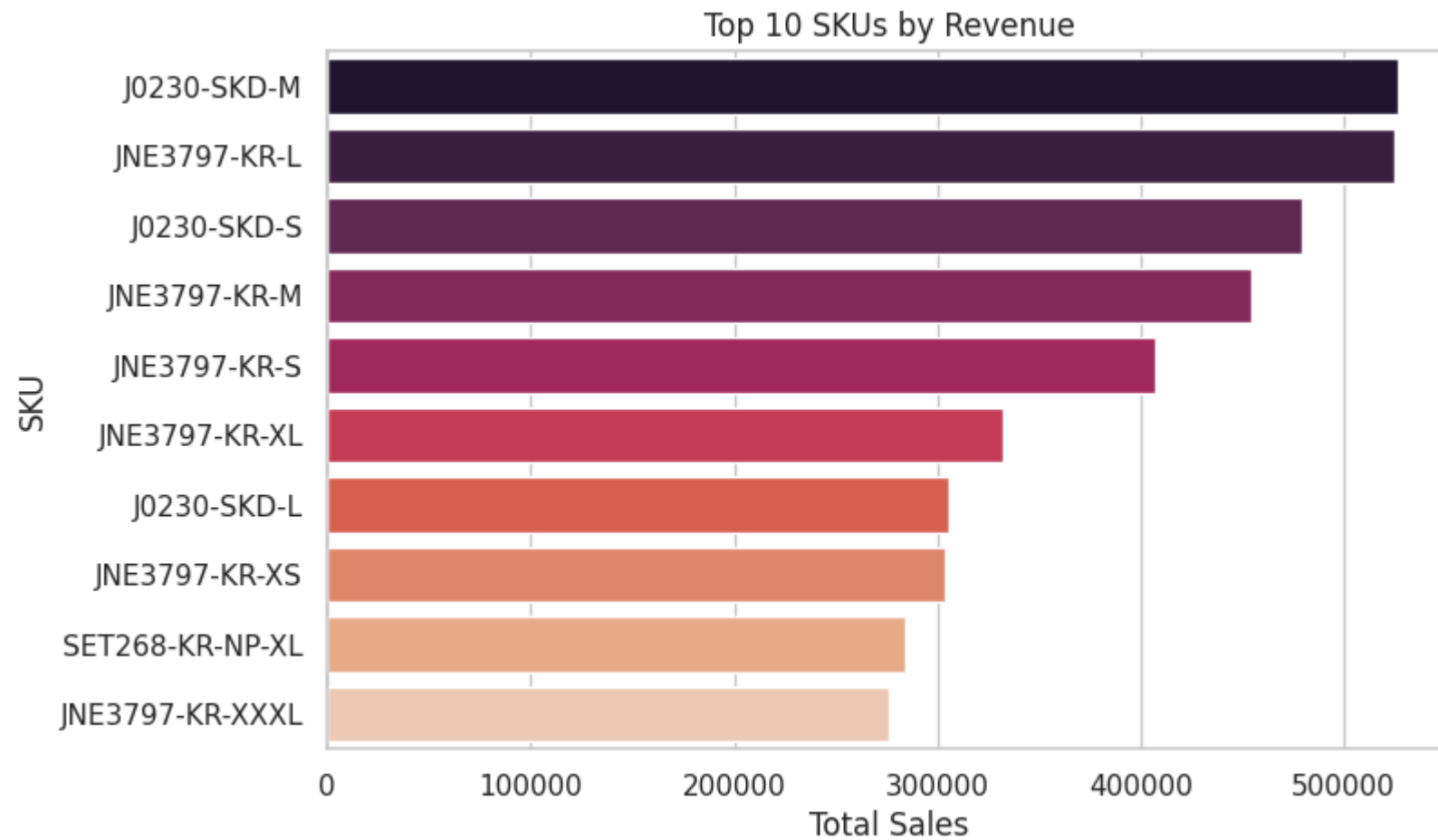
Top Products Analysis

Top 10 SKUs by Revenue

```
In [37]: top_sku = df.groupby("SKU")["Amount"].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(8,5))
sns.barplot(
    x=top_sku.values,
    y=top_sku.index,
    palette="rocket"
)

plt.title("Top 10 SKUs by Revenue")
plt.xlabel("Total Sales")
plt.show()
```

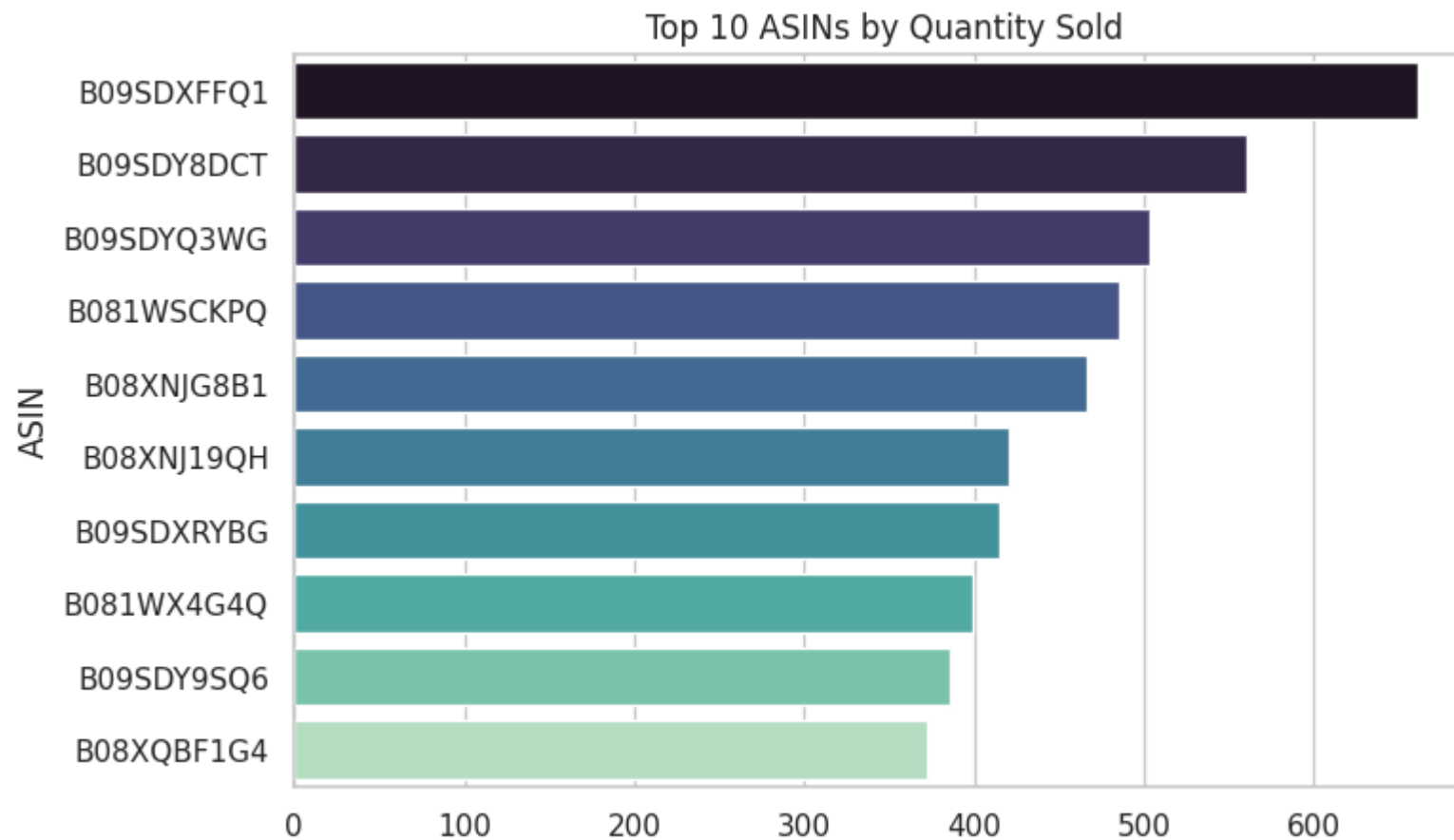


Top 10 ASINs by Quantity Sold

```
In [38]: top_asin = df.groupby("ASIN")["Qty"].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(8,5))
sns.barplot(
    x=top_asin.values,
    y=top_asin.index,
    palette="mako"
)
```

```
plt.title("Top 10 ASINs by Quantity Sold")
plt.show()
```

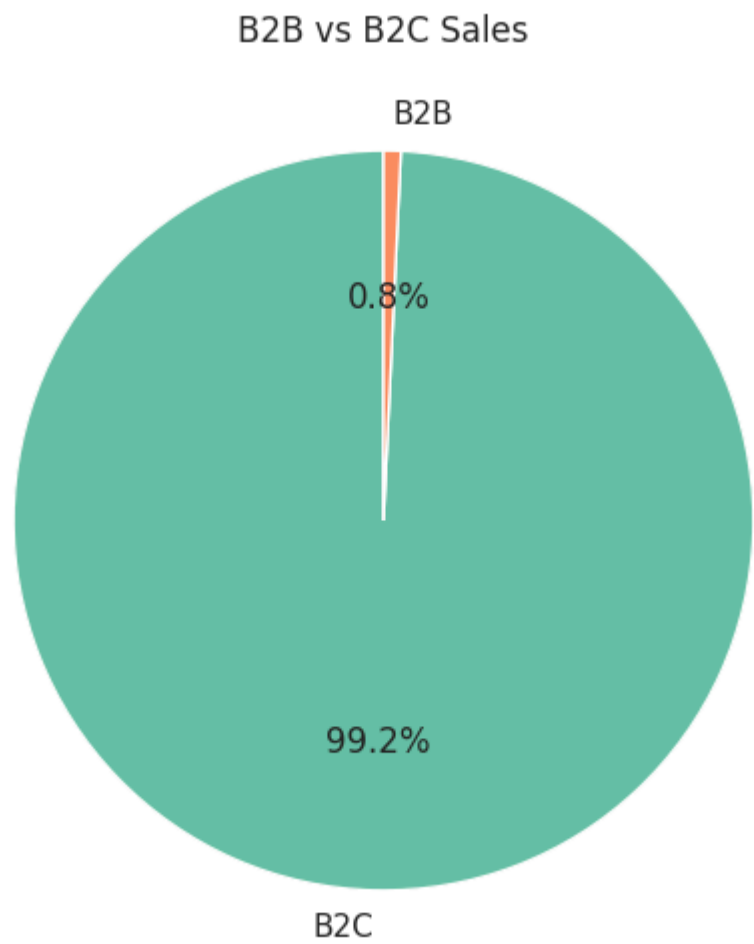


B2B vs B2C Sales

```
In [39]: b2b_sales = df.groupby("B2B")["Amount"].sum()

plt.figure(figsize=(6,6))
plt.pie(
    b2b_sales,
    labels=["B2C", "B2B"],
```

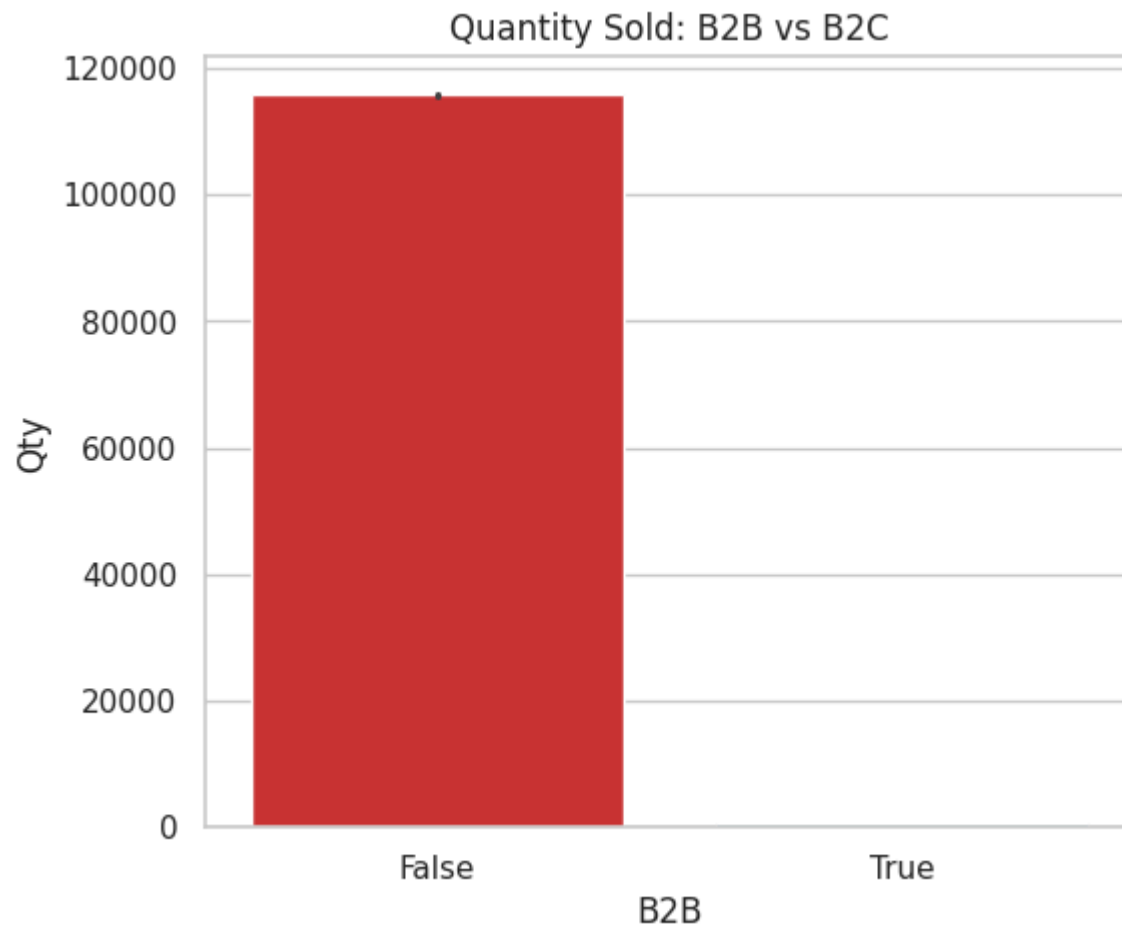
```
autopct="%1.1f%%",  
startangle=90,  
colors=["#66c2a5", "#fc8d62"]  
)  
  
plt.title("B2B vs B2C Sales")  
plt.show()
```



Quantity by B2B

```
In [40]: plt.figure(figsize=(6,5))
sns.barplot(
    x="B2B",
    y="Qty",
    data=df,
    estimator=sum,
    palette="Set1"
)

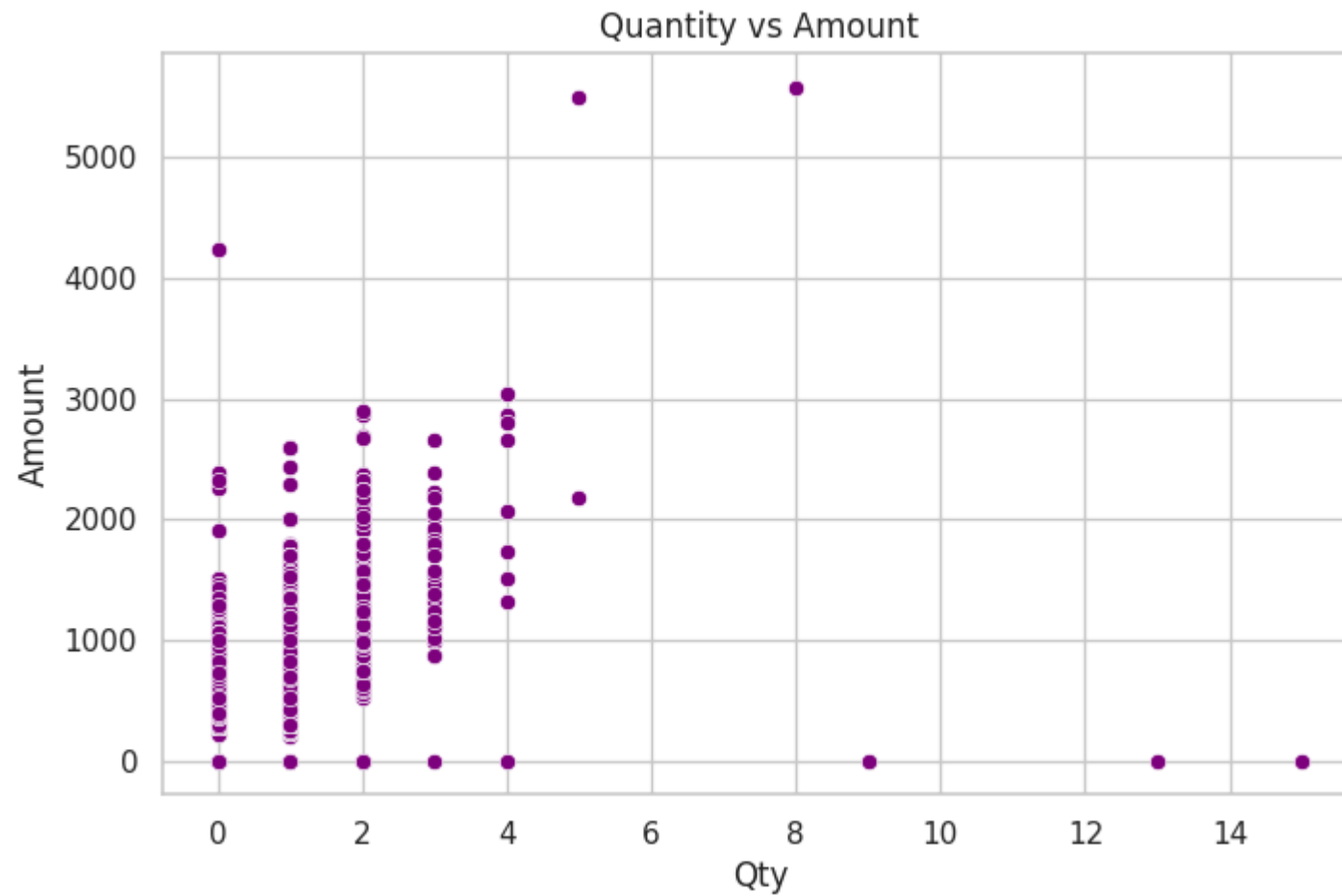
plt.title("Quantity Sold: B2B vs B2C")
plt.show()
```



Scatter: Quantity vs Amount

```
In [41]: plt.figure(figsize=(8,5))
sns.scatterplot(
    x="Qty",
    y="Amount",
    data=df,
    color="purple"
)
```

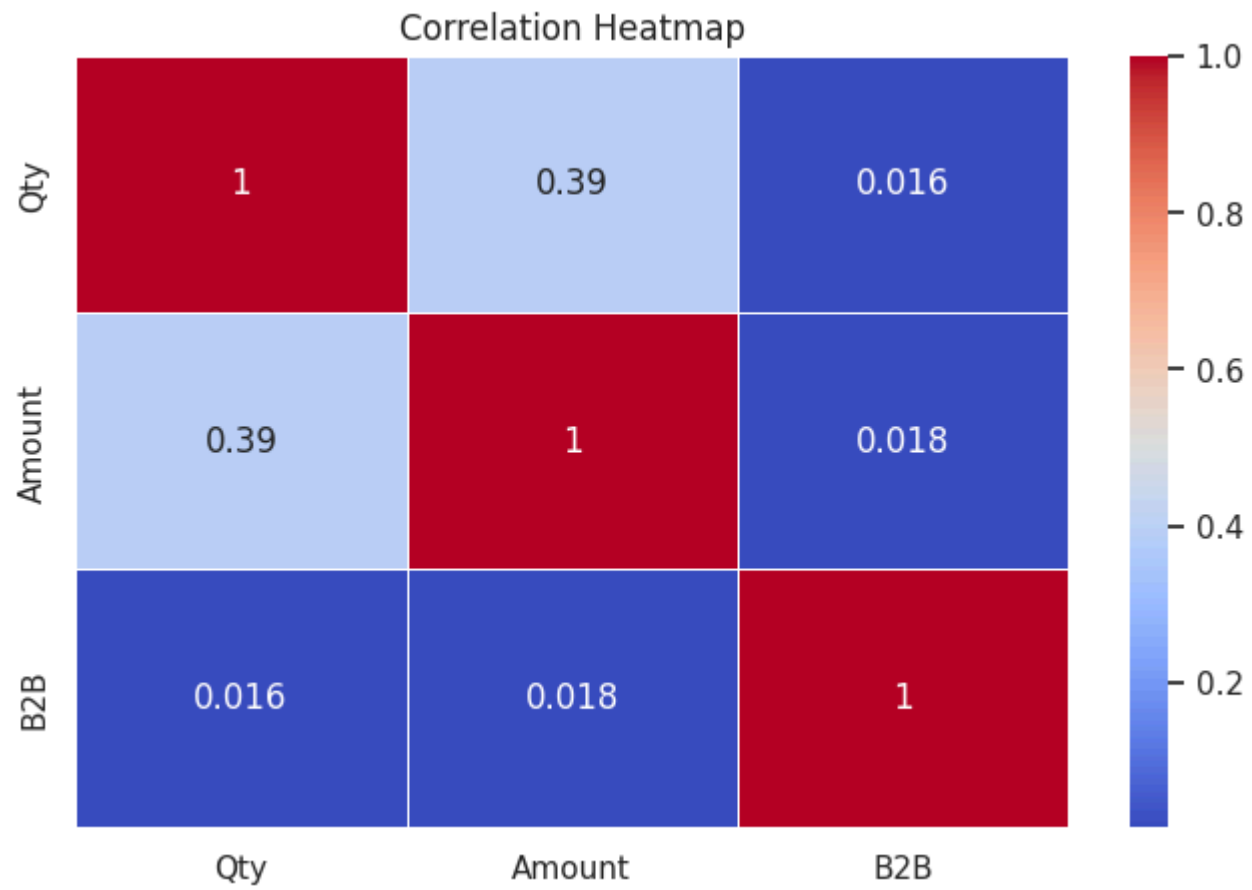
```
plt.title("Quantity vs Amount")  
plt.show()
```



Correlation Heatmap

```
In [42]: plt.figure(figsize=(8,5))  
sns.heatmap(  
    df[["Qty", "Amount", "B2B"]].corr(),  
    annot=True,  
    cmap="coolwarm",
```

```
linewidths=0.5
)  
plt.title("Correlation Heatmap")  
plt.show()
```



In []:

In []: